

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 7

End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

GitHub Repository

<https://github.com/MouryaAditya999/Deep-Learning-Lab/tree/main/Experiment7>

1 Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modeling. The experiment covers a Fully Connected Autoencoder (FC-AE), Convolutional Autoencoder (CAE), Denoising Convolutional Autoencoder, and Variational Autoencoder (VAE).

2 Learning Outcomes

After completing this experiment, the student is able to:

- explain the encoder–latent–decoder structure of an autoencoder;
- implement fully connected and convolutional autoencoders;
- introduce controlled noise and perform image denoising;
- evaluate reconstruction using MSE, MAE and SSIM;
- visualize latent representations;
- explain the probabilistic latent representation of a VAE;
- implement the reparameterization trick;
- interpret reconstruction and KL-divergence losses; and
- generate new images by sampling from a learned latent space.

3 Dataset and Experimental Setup

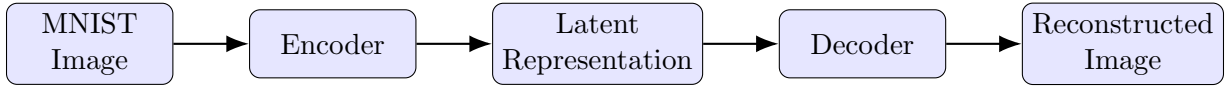
Dataset: MNIST Handwritten Digit Dataset.

Each image is a grayscale handwritten digit with dimensions $28 \times 28 \times 1$. The notebook uses 10,000 training images and 2,000 test images. The training portion is split into 9,000 training samples and 1,000 validation samples. Pixel values are normalized to the range $[0, 1]$.

Parameter	Value
Dataset	MNIST
Image size	$28 \times 28 \times 1$
Training images	9,000
Validation images	1,000
Test images	2,000
Pixel range	$[0, 1]$

Table 1: Experimental dataset setup

4 Overall Experimental Pipeline



5 Fully Connected Autoencoder

The fully connected autoencoder flattens each 28×28 image into a vector of length 784. The architecture used is:

$$784 \rightarrow 128 \rightarrow 32 \rightarrow 16 \rightarrow 32 \rightarrow 128 \rightarrow 784.$$

ReLU activation is used in hidden layers and sigmoid activation in the output layer.

Parameter	Value
Latent dimension	16
Optimizer	Adam
Learning rate	10^{-3}
Batch size	128
Epochs	20
Loss	Binary Cross-Entropy

Table 2: Fully connected autoencoder configuration

5.1 Training and Validation Results

The final training loss after 20 epochs was 0.1213 and the final validation loss was 0.1286.

Epoch	Training Loss	Validation Loss
1	0.3516	0.2606
5	0.1610	0.1613
10	0.1356	0.1405
15	0.1274	0.1335
20	0.1213	0.1286

Table 3: Selected FC-AE training and validation losses

Plot 1: Original vs Reconstructed (FC-AE)

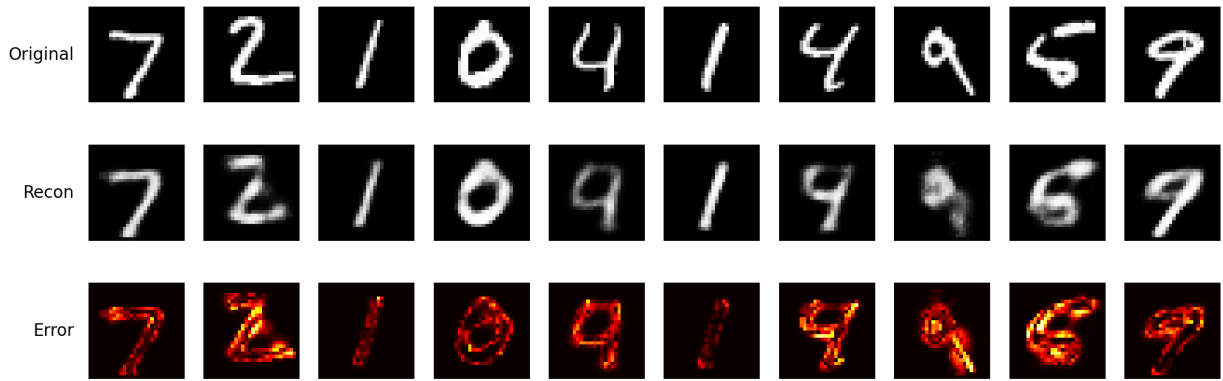


Figure 1: Original images and FC-AE reconstructions

Inference: The reconstructed images preserve the main digit shapes, while some fine details are lost because the input is compressed into a 16-dimensional latent representation.

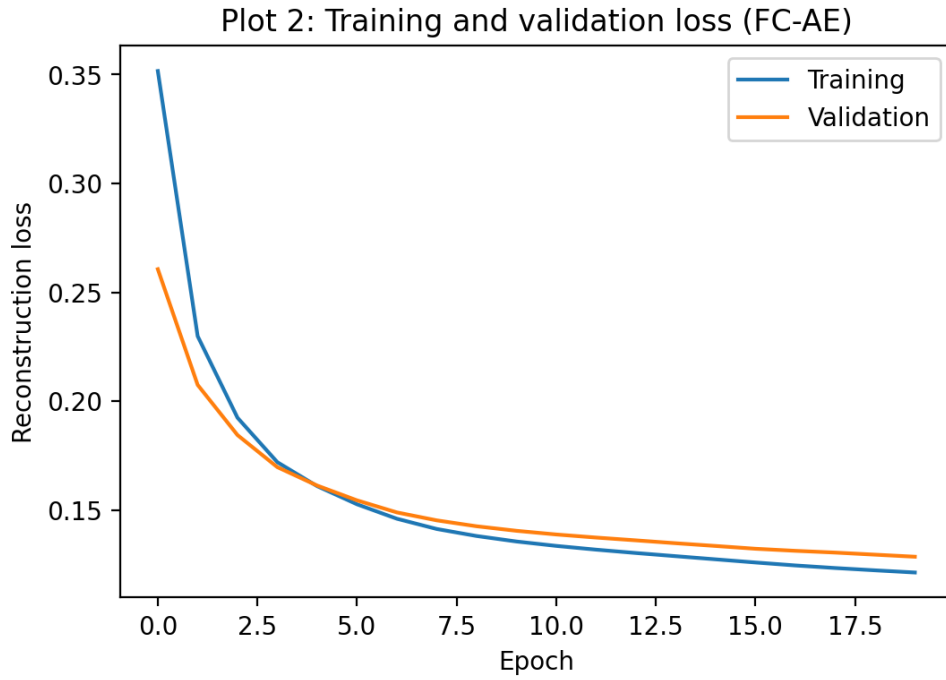


Figure 2: FC-AE training and validation reconstruction loss

Inference: Both losses decrease throughout training and remain close near the end, indicating convergence without a large validation-loss increase.

5.2 FC-AE Reconstruction Metrics

Metric	Value
Test MSE	0.02135
Test MAE	0.05662
Mean SSIM	0.74932
Parameters	211,040
Training Time (s)	32.95644

Table 4: FC-AE test-set results

6 Convolutional Autoencoder

The convolutional autoencoder preserves spatial information using convolution and pooling operations. The decoder uses upsampling and convolution operations.

$28 \times 28 \times 1 \rightarrow \text{Conv}(32) \rightarrow \text{Pool} \rightarrow \text{Conv}(64) \rightarrow \text{Pool} \rightarrow \text{Conv}(64) \rightarrow \text{Upsample} \rightarrow \text{Conv}(32) \rightarrow \text{Upsample} \rightarrow \text{Conv}(10)$

The CAE was trained for 20 epochs using the same MNIST subset.

The final CAE training loss was 0.0683 and the validation loss was 0.0694.

Plot 3: Original vs FC-AE vs CAE reconstruction

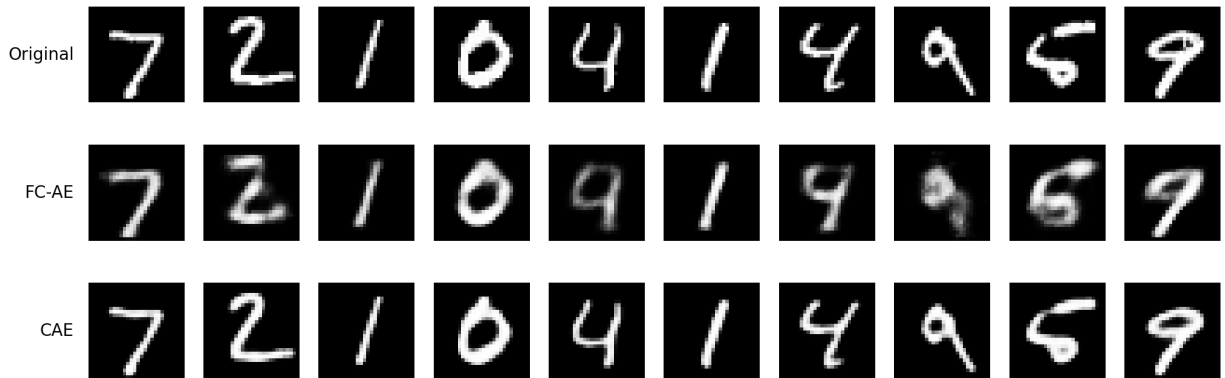


Figure 3: Original, FC-AE reconstruction and CAE reconstruction

Inference: The CAE reconstruction preserves spatial and local image structure more effectively than the fully connected reconstruction.

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.02135	0.05662	0.74932	211040
Convolutional AE	0.00258	0.01496	0.97423	74497

Table 5: FC-AE and CAE comparison

Inference: The CAE obtained lower MSE and MAE and higher SSIM than the FC-AE in this execution. It also used fewer trainable parameters.

7 Denoising Convolutional Autoencoder

A denoising autoencoder receives a corrupted image as input and uses the corresponding clean image as the target.

$$\tilde{x} \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}.$$

Two corruption types were evaluated: Gaussian noise and salt-and-pepper noise.

7.1 Gaussian Noise

A Gaussian-noise denoising CAE was trained with $\sigma = 0.2$.

Parameter	Value
Training noise	Gaussian
σ	0.2
Epochs	20
Final training loss	0.0759
Final validation loss	0.0781

Table 6: Gaussian-noise denoising training result

Plot 4: Clean vs Noisy vs Denoised (Gaussian, sigma=0.2)

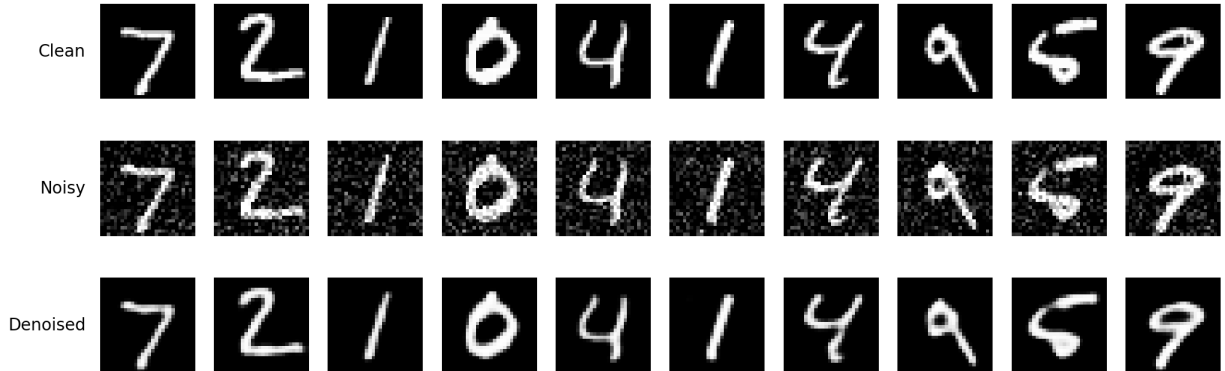


Figure 4: Clean, Gaussian-noisy and denoised images

Inference: The denoising CAE removes a substantial amount of Gaussian corruption while preserving the main structure of the handwritten digits.

7.2 Gaussian Noise-Level Study

Noise Level (σ)	MSE	MAE	SSIM
0.1	0.00378	0.01829	0.95614
0.2	0.00494	0.02200	0.94174
0.3	0.00755	0.02978	0.87907

Table 7: Gaussian noise level versus reconstruction quality

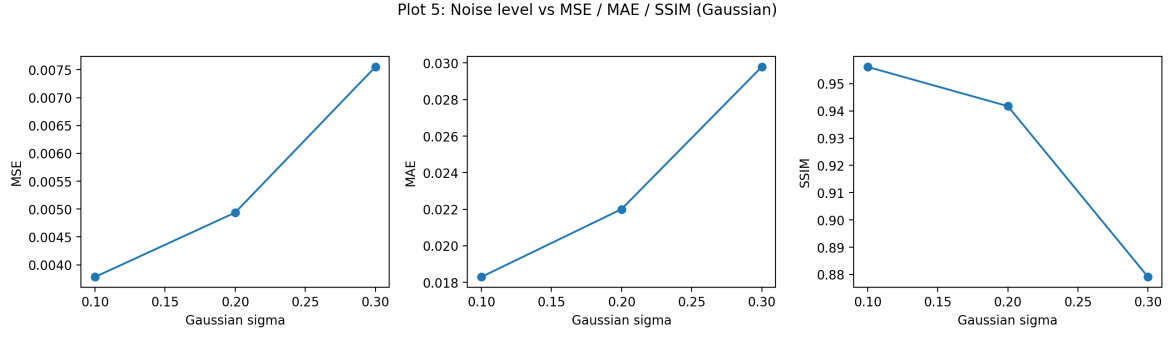


Figure 5: Gaussian noise level versus MSE, MAE and SSIM

Inference: Increasing Gaussian noise increases MSE and MAE while reducing SSIM. Therefore, stronger corruption makes accurate reconstruction more difficult.

7.3 Salt-and-Pepper Noise

A second denoising model was trained using salt-and-pepper noise with $p = 0.10$.

Noise Level (p)	MSE	MAE	SSIM
0.05	0.00406	0.01894	0.95700
0.10	0.00492	0.02114	0.94494
0.20	0.00804	0.02857	0.89202

Table 8: Salt-and-pepper noise level versus reconstruction quality

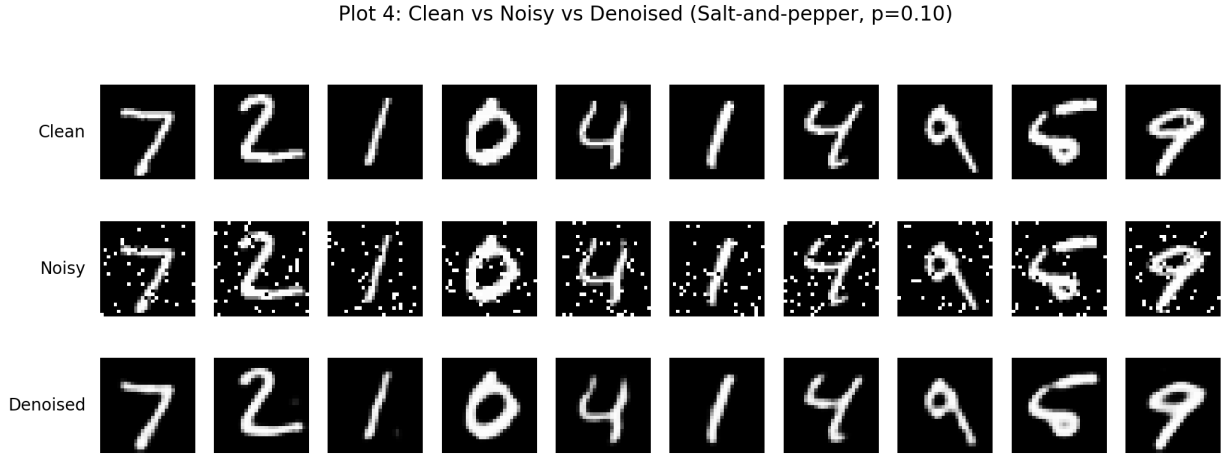


Figure 6: Clean, salt-and-pepper noisy and denoised images

Inference: The denoising model reduces salt-and-pepper corruption and recovers the major digit patterns. As the corruption probability increases, reconstruction error increases and SSIM decreases.

8 Variational Autoencoder

The VAE learns a probability distribution in the latent space instead of a single deterministic latent vector. The encoder produces μ and $\log \sigma^2$ and sampling is performed using

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

The VAE objective is

$$L_{\text{VAE}} = L_{\text{rec}} + L_{\text{KL}}.$$

The experiment uses a two-dimensional latent space.

8.1 VAE Training Results

The VAE was trained for 30 epochs. The final reported test metrics were:

Metric	Value
Reconstruction Loss	153.36845
KL Loss	5.69516
Total Loss	159.06362
Test MSE	0.04536
Test MAE	0.10687
Mean SSIM	0.46529
Parameters	134165
Training Time (s)	206.72349

Table 9: VAE results

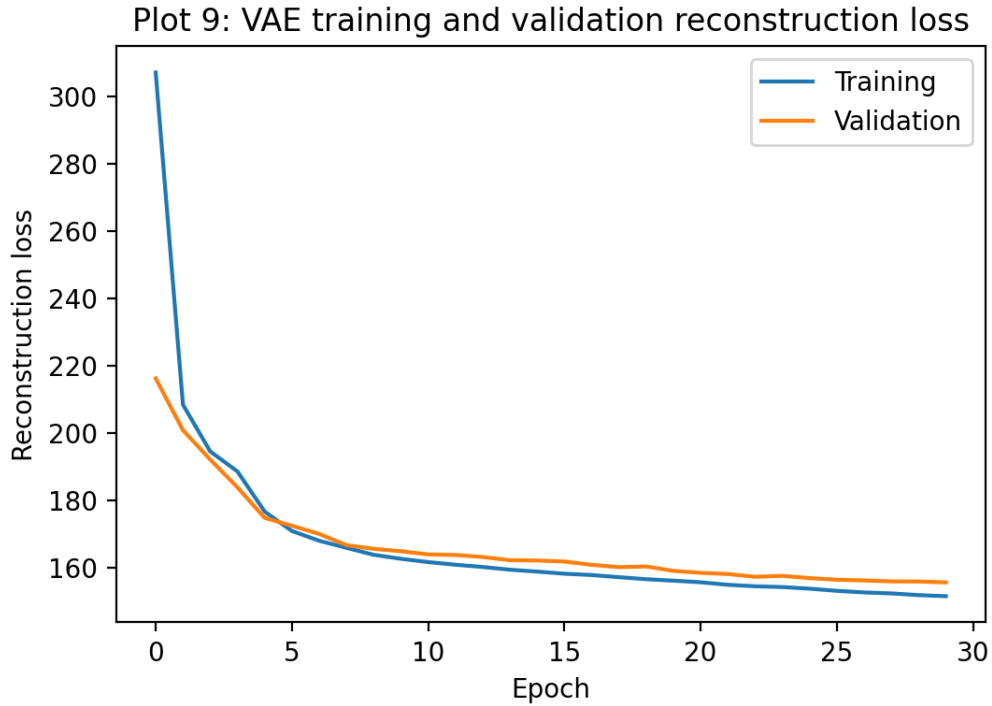


Figure 7: VAE training and validation reconstruction loss

Inference: The reconstruction loss decreases over training and stabilizes, showing that the VAE learns to reconstruct the MNIST images while also maintaining its probabilistic latent representation.

8.2 VAE Latent-Space Visualization

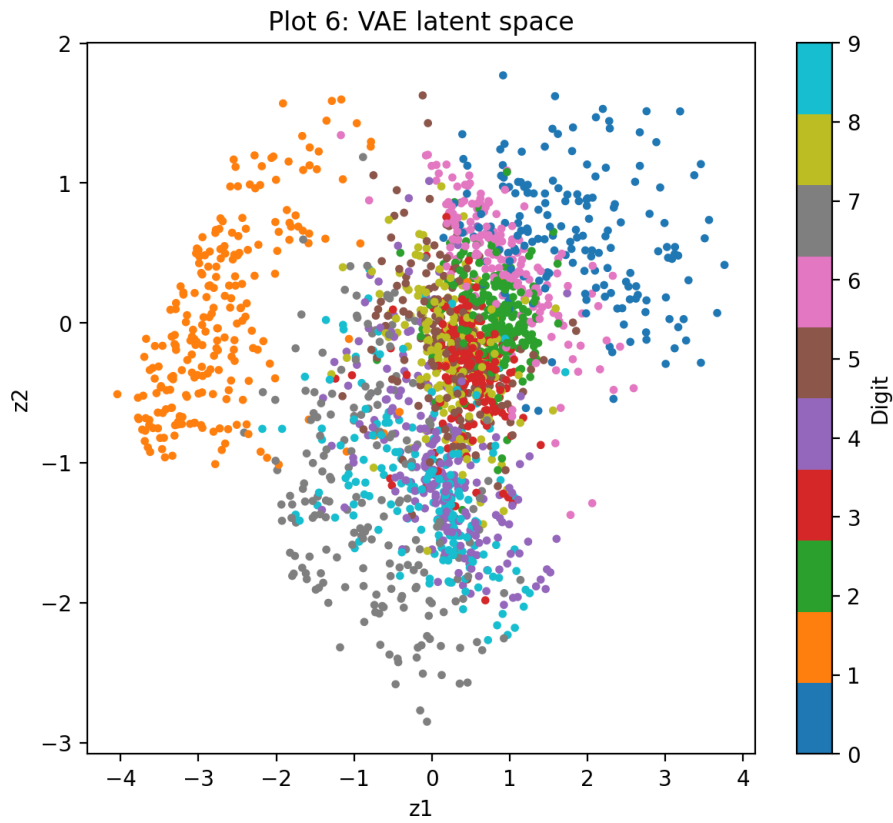


Figure 8: Two-dimensional VAE latent space

Inference: The two-dimensional latent space organizes digit samples into regions with partial class separation, while some classes overlap because the VAE learns a continuous probabilistic representation.

8.3 Random VAE Generation

Plot 7: Randomly generated VAE images



Figure 9: 25 randomly generated VAE images

Inference: Random samples drawn from the learned latent prior produce digit-like images with visible variation. Some generated samples may be less distinct because the latent space is continuous and the model balances reconstruction with KL regularization.

8.4 Latent-Space Interpolation

Two latent points corresponding to different digit samples were interpolated using

$$z(\alpha) = (1 - \alpha)z_A + \alpha z_B, \quad \alpha \in \{0, 0.1, \dots, 1.0\}.$$

Plot 8: Latent space interpolation

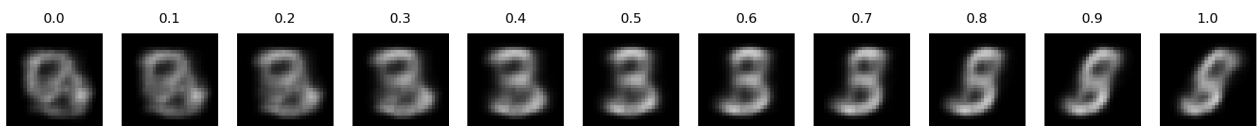


Figure 10: Latent-space interpolation

Inference: The decoded images change gradually as the latent vector moves between the two endpoints, demonstrating the continuous nature of the VAE latent space.

9 Reconstruction Error Distribution

The per-image reconstruction error was calculated using the CAE:

$$e_i = \frac{1}{D} \sum_{j=1}^D (x_{ij} - \hat{x}_{ij})^2.$$

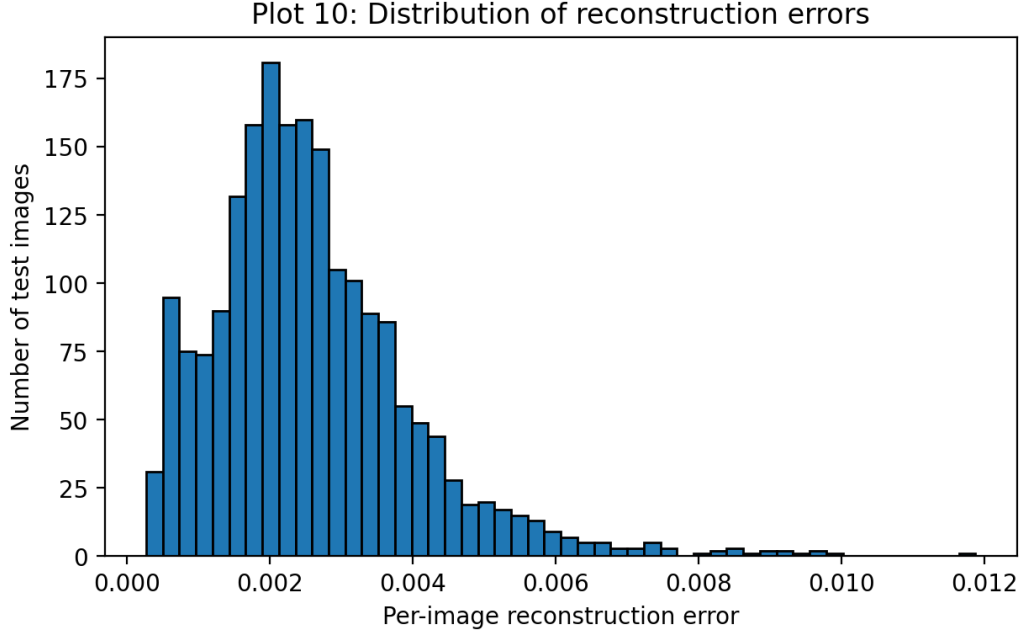


Figure 11: Distribution of CAE reconstruction errors on the test set

Statistic	Value
Count	2000
Mean	0.00258
Standard deviation	0.00141
Minimum	0.00027
25th percentile	0.00167
Median	0.00236
75th percentile	0.00326
Maximum	0.01188

Table 10: CAE reconstruction-error statistics

Inference: Most test images have relatively low reconstruction error, while a smaller number of samples have noticeably larger errors.

9.1 Five Highest-Error Samples

Five test images with the largest reconstruction error



Figure 12: Five test images with the largest CAE reconstruction errors

Inference: The highest-error samples show larger differences between the original and reconstructed images. Such samples can contain unusual writing styles or digit structures that are harder for the model to represent.

10 Latent-Dimension Study

The fully connected autoencoder was trained with latent dimensions 2, 8, 16 and 32.

Latent Dimension	MSE	SSIM	Parameters
2	0.04561	0.45415	210130
8	0.02648	0.68745	210520
16	0.02309	0.72429	211040
32	0.02091	0.75174	212080

Table 11: Effect of latent dimension on FC-AE reconstruction

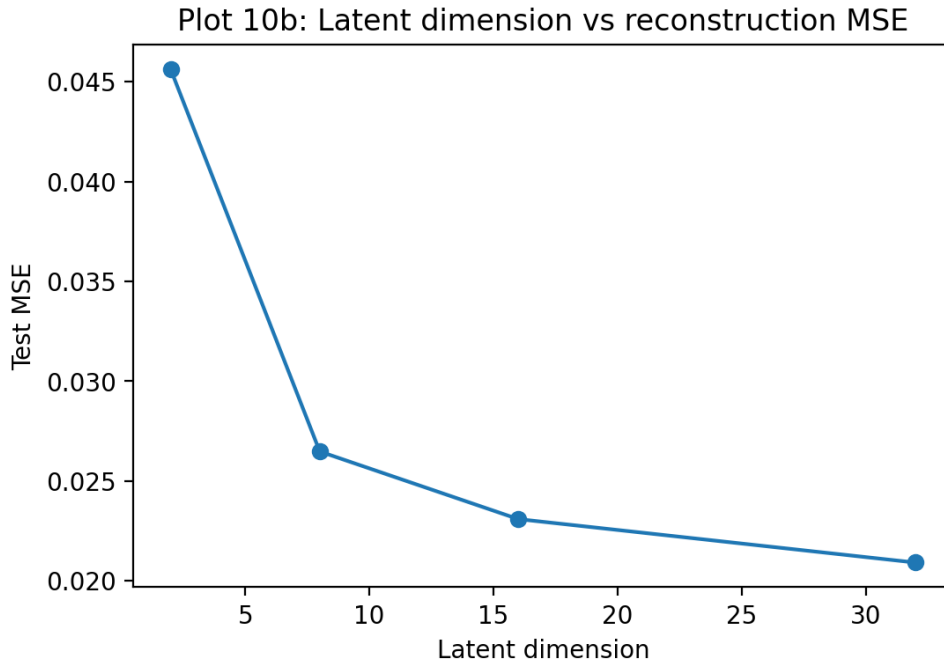


Figure 13: Latent dimension versus reconstruction MSE

Inference: Increasing the latent dimension reduces MSE and improves SSIM because the model can retain more information about the input image. The trade-off is reduced compression and a small increase in the number of parameters.

11 Consolidated Results

The following values were obtained directly from the supplied notebook execution.

Model	MSE	MAE	SSIM	Parameters	Time (s)
FC Autoencoder	0.02135	0.05662	0.74932	211040	32.95644
Conv. Autoencoder	0.00258	0.01496	0.97423	74497	190.40687
Denoising CAE	0.00494	0.02200	0.94174	74497	166.64772
VAE	0.04536	0.10687	0.46529	134165	206.72349

Table 12: Consolidated experimental results

Observation: In this execution, the CAE produced the lowest reconstruction errors and highest SSIM among the reconstruction-based models. The denoising CAE also achieved strong reconstruction quality while operating on corrupted inputs. The VAE has a different objective because its total loss includes both reconstruction and KL-divergence terms.

12 Discussion Questions

1. What is an autoencoder?

An autoencoder is a neural network that learns to reconstruct its input using an encoder, latent representation and decoder.

2. What is the purpose of the encoder?

It converts the input into a compact latent representation.

3. What is the purpose of the decoder?

It reconstructs the input from the latent representation.

4. **What is a latent representation?**
It is a compact representation containing important information from the input.
5. **Why is the latent representation usually smaller than the input?**
To force the model to learn the most important features.
6. **What happens when the latent dimension is too small?**
Important information is lost and reconstruction quality decreases.
7. **What happens when the latent dimension is increased?**
More information can be retained, generally improving reconstruction.
8. **Why can a convolutional autoencoder preserve spatial information more effectively?**
Convolutional layers process local regions while preserving spatial relationships.
9. **Differentiate an autoencoder and a denoising autoencoder.**
An autoencoder reconstructs the input, while a denoising autoencoder reconstructs a clean image from a corrupted input.
10. **Why is the clean image used as the target in denoising?**
It teaches the network to remove the introduced corruption.
11. **What happens when the noise level is increased?**
Reconstruction becomes harder, so MSE and MAE generally increase while SSIM decreases.
12. **Why are MSE, MAE and SSIM useful for image reconstruction?**
MSE and MAE measure pixel-level errors, while SSIM measures structural similarity.
13. **What is a Variational Autoencoder?**
A VAE learns a probability distribution in the latent space and can generate new samples.
14. **How does a VAE differ from a conventional autoencoder?**
A conventional autoencoder uses a deterministic latent vector, while a VAE learns a latent probability distribution.
15. **Why does a VAE learn a distribution instead of a single latent vector?**
A continuous distribution enables meaningful sampling and generation.
16. **What is the reparameterization trick?**
It represents sampling as $z = \mu + \sigma \odot \epsilon$ so gradients can pass through the sampling operation.
17. **Why is ϵ sampled from a standard normal distribution?**
It provides controlled randomness while μ and σ remain learnable.
18. **What is the role of KL divergence in the VAE loss?**
It encourages the learned latent distribution to remain close to the prior.
19. **What is the prior distribution used in the VAE?**
Usually $\mathcal{N}(0, I)$.
20. **Why is a two-dimensional latent space useful for visualization?**
It can be directly plotted and inspected.
21. **What does latent-space interpolation demonstrate?**
It demonstrates whether the decoder produces smooth transitions between latent representations.
22. **Why can a VAE generate new images?**
Latent vectors sampled from the learned prior can be passed through the decoder.
23. **Why might some VAE-generated images be unclear?**
Limited model capacity, training effects and sampling from less representative latent regions can produce unclear outputs.
24. **What is the difference between reconstruction and generation?**
Reconstruction reproduces a given input, while generation creates a new sample from a latent vector.
25. **Why should test images remain unseen during training?**
They provide an unbiased measure of how well the model generalizes to unseen data.

13 Overall Discussion

The experiment demonstrates the progression from deterministic reconstruction to convolution-based image representation, denoising and probabilistic generation. The FC-AE achieved a test MSE of 0.02135 and mean SSIM of 0.74932. The CAE improved the reconstruction metrics to an MSE of

0.00258 and SSIM of 0.97423 while using fewer parameters.

The denoising CAE achieved an MSE of 0.00494, MAE of 0.02200 and SSIM of 0.94174 at Gaussian noise level $\sigma = 0.2$. The noise-level study showed decreasing reconstruction quality as corruption increased.

The VAE learned a two-dimensional probabilistic latent representation. Its test MSE was 0.04536 and mean SSIM was 0.46529, while its total objective included a reconstruction loss of 153.36845 and KL loss of 5.69516. The latent visualization, random generation and interpolation demonstrate the generative properties of the VAE.

The latent-dimension experiment also showed that increasing the FC-AE latent dimension from 2 to 32 reduced MSE from 0.04561 to 0.02091 and increased SSIM from 0.45415 to 0.75174.

14 Conclusion

The experiment successfully implemented and evaluated fully connected, convolutional, denoising and variational autoencoders on MNIST. Quantitative metrics and visualizations were used to study reconstruction quality, denoising performance, latent representations and generative behavior. The results demonstrate the importance of spatial structure in convolutional autoencoders, the effect of corruption level on denoising quality, and the probabilistic nature of VAE latent spaces.

15 References

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
2. Diederik P. Kingma and Max Welling, “Auto-Encoding Variational Bayes,” ICLR, 2014.
3. Pascal Vincent et al., “Stacked Denoising Autoencoders: Learning Useful Representations in a Deep Network with a Local Denoising Criterion,” Journal of Machine Learning Research, 2010.
4. Yann LeCun, Corinna Cortes and Christopher J. C. Burges, “MNIST Handwritten Digit Database.”
5. TensorFlow Documentation: <https://www.tensorflow.org>
6. Keras Documentation: <https://keras.io>
7. scikit-image Documentation: <https://scikit-image.org>